

Safe Observable Orbital Arc Planning for Full-Coverage Autonomous Inspection of Orbital Structures

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Abstract— Autonomous inspection of large orbital structures requires inspection actions that are simultaneously informative, dynamically reachable, sensor-valid, and safe under relative orbital motion. This paper introduces a safe observable orbital arc (SOOA) representation for full-coverage orbital inspection. Each SOOA is an inspection motion primitive that jointly encodes motion governed by the Hill–Clohessy–Wiltshire (HCW) model, camera field-of-view visibility, line-of-sight surface coverage, transfer effort, clearance, input feasibility, and a passive-safety audit. The inspection problem is formulated as constrained SOOA selection rather than geometric viewpoint selection followed by dynamic feasibility checking. For International Space Station (ISS)-scale mesh instances, a scalable deterministic solver uses these visibility and feasibility semantics to generate high-coverage safe inspection tours, while a small exact dynamic-programming oracle is retained only to check finite-graph optimality on exhausted reduced SOOA graphs. The method is evaluated on an open National Aeronautics and Space Administration (NASA) ISS mesh against coverage-greedy, nearest-view, fuel-focused, Clohessy–Wiltshire next-best-view (CW-NBV), random-safe, and safety-filtered coverage baselines. The proposed planner reaches 98.33% weighted surface-sample coverage with 21 selected inspection actions, 21.592 m/s cumulative Δv , 9.91 m minimum clearance, 1.51 m minimum passive-drift audit margin over a 300 s horizon, and 5.63 s planning time in the reported base case. Baselines that optimize coverage geometrically can reach similar coverage only with input or clearance violations, whereas feasible safe baselines require substantially higher effort. Robot Operating System 2 (ROS 2), RViz, and Gazebo replay verify that the logged trajectory, attitude stream, and camera frusta are consistent with the offline SOOA visibility model, while all quantitative dynamics, coverage, and safety claims remain based on the HCW planner logs.

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Index Terms— spacecraft inspection, relative orbital dynamics, safe observable orbital arc, field-of-view constraints, coverage planning, passive safety, ROS verification, Hill–Clohessy–Wiltshire equations

I. Introduction

Large orbital structures are becoming central to civil, scientific, and commercial spaceflight. Space stations, assembled telescopes, logistics platforms, and in-space manufacturing facilities all require periodic external inspection to detect configuration changes, micrometeoroid damage, thermal blanket degradation, docking-interface anomalies, and other faults that are not easily observed from fixed onboard cameras. The corresponding planning problem is simultaneously geometric and dynamic. A candidate view may provide high visual coverage yet remain unsafe, expensive, or unreachable under relative orbital motion.

Existing inspection planners often emphasize one side of this problem. Robotic next-best-view methods provide mature tools for coverage, occlusion reasoning, and sensor placement, but their motion cost is usually geometric. Relative orbit-design methods provide dynamically meaningful paths, but they do not always treat surface coverage, camera field of view, and target occlusion as the planning objective. This separation is limiting for large structures. Selecting views first and checking dynamics later can produce trajectories that appear coverage-rich but violate clearance or require excessive input.

The key difficulty is not only that coverage and dynamics must both be considered, but that the selectable inspection action itself must be physically meaningful. A static viewpoint is an incomplete planning primitive for orbital inspection because a view that is geometrically informative may be unreachable, unsafe during approach, or passively unsafe after thrust loss. Conversely, an orbit segment that is dynamically natural may be uninformative if its camera field of view does not expose new target surface. This motivates an inspection primitive that binds observability and orbital feasibility before planning begins.

This paper addresses that gap by introducing the safe observable orbital arc (SOOA). A SOOA is not merely a candidate viewpoint or a transfer edge. It is a finite-duration inspection action generated under relative orbital dynamics and annotated with the surface targets observed through the camera model, the required control effort, the minimum clearance, the input and speed feasibility status, and a passive-drift safety audit. Planning over SOOAs changes the finite search space: infeasible or sensor-invalid actions are removed before coverage selection, rather than repaired afterward. In the present implementation, geometric viewpoints remain as seeds for transfer-to-dwell arcs, but the exported and evaluated planning object is the selected SOOA.

The finite graph also gives a precise scope for verification. For a fixed SOOA library, fixed visibility relation,

and fixed feasible-transition graph, the inspection task can be written as a mixed-integer optimization problem. We retain an exact dynamic-programming oracle on small exhausted graphs to verify the discrete SOOA objective. This oracle is not presented as the full-mission solver; the reported ISS-scale high-coverage trajectories are generated by the scalable deterministic SOOA planner.

The reported results show that dynamics awareness matters. On the ISS-mesh benchmark, the proposed SOOA tour achieves 98.33% coverage with 21 selected inspection actions, 21.592 m/s cumulative Δv , 9.91 m minimum clearance, and 5.97 s planning time. Coverage-greedy reaches the same coverage but requires input beyond the admissible bound and violates clearance; nearest selection also reaches high coverage but violates clearance. Safe-coverage and random-safe baselines remain feasible at high coverage, but require substantially higher cumulative Δv , while the fuel-focused baseline stops below the high-coverage target. The comparison therefore exposes failure modes that are hidden by coverage metrics alone in this deterministic base case. Robustness across target density, candidate density, initial state, keep-out margin, and transfer duration is treated as a separate validation requirement rather than inferred from one mesh instance.

ROS 2 Jazzy and Gazebo Harmonic are used as a replay and verification layer, not as the source of orbital dynamics. The ROS stack publishes the chaser state from the HCW dynamics model, records trajectory, attitude, coverage, safety, and planner logs, and reconstructs the camera frustum used by the offline planner. Gazebo provides visual context for the station and chaser. This replay layer verifies that the planned visual geometry can be instantiated in the integrated system while preserving the planner as the scientific object of study.

The main contributions are:

- 1) We introduce the safe observable orbital arc (SOOA), an inspection motion primitive that jointly encodes HCW relative-motion feasibility, FOV-valid surface visibility, line-of-sight coverage, control effort, clearance, input feasibility, and passive-safety audit metrics.
- 2) We formulate full-coverage orbital inspection as a constrained SOOA-selection problem on a finite graph, avoiding the conventional separation between geometric viewpoint selection and subsequent dynamic feasibility checking.
- 3) We develop a scalable deterministic safe-arc coverage solver for ISS-scale mesh benchmarks, with an exact dynamic-programming oracle used only to verify reduced exhausted SOOA graphs.
- 4) We provide a reproducible validation protocol with baseline comparisons, sensitivity studies, ablations, and ROS 2/RViz/Gazebo replay that verifies trajectory, attitude, camera-frustum, coverage, and safety-log consistency while preserving HCW propagation as the quantitative source of truth.

II. Related Work

Orbital inspection draws on relative spacecraft motion, robotic view planning, and orbit-design methods for complex structures. The Clohessy–Wiltshire equations remain a common starting point for rendezvous and proximity operations near a circular chief orbit because they provide a linear, interpretable model of relative translation [1]. Although high-fidelity perturbation models are required for flight design, a CW/HCW model is appropriate for the planning question studied here: how view ordering affects fuel use, clearance, and input demand under a fixed relative dynamics model [2].

Robotic inspection and next-best-view planning study how a robot should select sensor poses that reconstruct or inspect an object efficiently. Classical surveys emphasize visibility, occlusion, sensor range, view quality, and the trade-off between coverage and motion cost [3]. Kinodynamic sampling-based planners, including rapidly exploring random trees and their optimal variants, explicitly combine dynamics and collision constraints during motion planning [4], [5]. Their strength is continuous-state reachability, whereas coverage is usually encoded through task-specific rewards, terminal goals, or repeated replanning rather than as a finite weighted dwell-coverage objective over a spacecraft surface.

Recent spacecraft-inspection studies illustrate the same tension from different directions. Apodaca et al. couple viewpoint planning with trajectory generation for in-orbit structure inspection and emphasize coverage-energy trade-offs in a supervised planning pipeline [6]. Hou et al. design relative orbits for visual inspection of the China Space Station and analyze field-of-view and occlusion feasibility through orbital geometry [7]. Zhou et al. plan tomographic inspection paths around complex spacecraft structures with surface-aware obstacle avoidance [8]. Ogri et al. address a complementary on-line problem: image-based feature tracking, information gain, and dwell-time conditions for autonomous inspection under switched systems [9]. The present work is positioned between these lines: it does not introduce a new orbital dynamics model, a flight-certified collision-avoidance controller, or a globally optimal continuous planner. Its contribution is the SOOA abstraction and offline finite-arc solver in which surface coverage, camera feasibility, HCW transfer effort, clearance, input saturation, and passive-drift audit metrics are evaluated before an inspection action can contribute coverage.

The distinction is the planning primitive. Robotic NBV and inspection planners commonly reason over sensor poses or short geometric motions, while relative-orbit design methods reason over dynamically meaningful paths that are not necessarily coverage-optimal. The proposed SOOA representation merges these two views at the discrete action level: every selectable action is simultaneously an orbital motion segment and a sensor-valid coverage contributor. This makes the finite graph a

TABLE I: Positioning relative to closely related inspection and planning approaches.

Approach	Primary emphasis	Difference from this work
Robotic NBV surveys [3]	geometric surface or volume coverage with sensor-pose sampling	motion cost is usually geometric or platform-specific rather than an HCW-feasible orbital edge cost
Kinodynamic sampling [4], [5]	continuous-state motion planning with dynamics and collision constraints	coverage is typically task-reward driven rather than a finite weighted dwell-coverage graph over spacecraft surface targets
Apodaca et al. [6]	inspection coverage and energy trade-off for in-orbit structures	does not make the selectable primitive a safety-audited observable orbital arc with reduced-graph oracle verification
Hou et al. [7]	relative-orbit design for visual inspection with FOV and occlusion checks	optimizes orbit geometry rather than selecting finite-duration SOOAs for weighted surface coverage
Zhou et al. [8]	tomographic inspection paths around complex spacecraft structures	targets scan-path planning rather than HCW-feasible dwell-view sequencing with input auditing
Ogri et al. [9]	online feature tracking, information gain, and switched-system execution	addresses execution-level camera control rather than offline finite-graph coverage planning
This work	safe observable orbital arcs for FOV-valid weighted surface coverage	selects finite-duration HCW-feasible inspection actions annotated with visibility, fuel cost, clearance, input feasibility, passive-safety audit, reduced-graph oracle verification, and ROS 2/RViz/Gazebo replay

graph of admissible inspection actions, not merely a graph of geometric views connected by post hoc transfers.

Table I clarifies what is gained and lost by the finite SOOA graph. The gain is a common discrete object on which coverage, camera feasibility, transition feasibility, clearance, input limits, passive-drift audit metrics, and reduced-graph oracle checks can be stated simultaneously. The cost is that optimality is conditional on the sampled SOOA library and precomputed feasible-transition graph; the approach does not provide global optimality in continuous viewpoint space, nor does it replace kinodynamic planning when a mission requires continuous online obstacle avoidance. The safety statement is therefore planning-level feasibility filtering and audit on logged HCW rollouts, not a flight safety certificate.

Simulation frameworks for space robotics must also be treated carefully. A rendered Gazebo scene can provide visual context and reveal integration errors in camera geometry, but standard rigid-body physics in a robotics simulator is not a substitute for orbital relative dynamics. We therefore use Gazebo Harmonic for station and chaser visualization while the ROS 2 dynamics node publishes the state used by replay, safety checks, camera-frustum verification, and logging. This separation makes the validation repeatable without allowing simulator physics to define the scientific result.

III. System Model and Mission Setting

The mission is divided into an offline planning problem and an execution verification problem. Offline planning receives the station geometry, initial relative state, camera parameters, keep-out distance, and actuation limits. It returns a time-indexed reference trajectory, a camera-attitude stream, and summary metrics for coverage, effort, clearance, and feasibility. Execution then replays or tracks this reference and checks that the logged state, attitude, and camera frustum are consistent with

the planning model. The planner is therefore evaluated by HCW-based trajectory and camera outputs, whereas ROS 2 and Gazebo verify integration and visual geometry.

A. Reference Frame and Dynamics

The inspection pass is represented in a local-vertical/local-horizontal frame attached to a nominal circular chief orbit. Over the time horizon considered here, the inspected structure is treated as fixed in this frame. The chaser translational state is

$$\mathbf{x}(t) = [\mathbf{r}^T(t) \quad \mathbf{v}^T(t)]^T \in \mathbb{R}^6, \quad (1)$$

where $\mathbf{r} = [r_x, r_y, r_z]^T \in \mathbb{R}^3$ and $\mathbf{v} = [v_x, v_y, v_z]^T \in \mathbb{R}^3$ are the relative position and velocity, respectively. The camera boresight is modeled through an attitude variable or an equivalent unit pointing vector. In the baseline mission, attitude enters the visibility predicate and the replay stream, while closed-loop attitude dynamics are left outside the offline optimization. This scope isolates the coupling between surface coverage and translational orbital motion.

The baseline continuous-time dynamics are the Hill–Clohessy–Wiltshire (HCW) equations,

$$\dot{\mathbf{r}} = \mathbf{v}, \quad (2)$$

$$\dot{\mathbf{v}} = \begin{bmatrix} 3n^2 r_x + 2nv_y + a_x \\ -2nv_x + a_y \\ -n^2 r_z + a_z \end{bmatrix}, \quad (3)$$

where n is the chief mean motion and $\mathbf{u} = [a_x, a_y, a_z]^T \in \mathbb{R}^3$ is the commanded acceleration. Equivalently, these dynamics have the linear state-space form

$$\dot{\mathbf{x}}(t) = A_{\text{HCW}}\mathbf{x}(t) + B\mathbf{u}(t), \quad (4)$$

with

$$A_{\text{HCW}} = \begin{bmatrix} 0_{3 \times 3} & I_3 \\ K_n & C_n \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3 \times 3} \\ I_3 \end{bmatrix}, \quad (5)$$

$$K_n = \text{diag}(3n^2, 0, -n^2), \quad C_n = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (6)$$

Here, $0_{3 \times 3}$ and I_3 are the 3×3 zero and identity matrices, respectively. For planning and numerical propagation, the input is assumed constant over each sampling interval of duration Δt . The resulting exact zero-order-hold discrete model is

$$\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k, \quad (7)$$

$$A_d = e^{A_{\text{HCW}} \Delta t}, \quad (8)$$

$$B_d = \int_0^{\Delta t} e^{A_{\text{HCW}} \tau} B d\tau. \quad (9)$$

where $\mathbf{x}_k = \mathbf{x}(k\Delta t)$, k is the discrete-time index, and $\mathbf{u}_k$ is the commanded acceleration over the k th interval. This discrete HCW model is used in both the offline trajectory generator and the ROS dynamics node. Gazebo is not used to propagate the spacecraft state in the reported experiments.

B. Inspection Geometry and Termination

The inspected structure is represented for planning by a finite set $\mathcal{T} = \{1, \dots, N\}$ of target samples. Each target i has a point p_i , outward normal n_i , and nonnegative weight w_i proportional to the surface area represented by that sample. A target is considered inspected when it has remained visible for at least $\tau_{\min}$. The baseline formulation does not include solar illumination dynamics; if required by a mission scenario, illumination can be added as a binary feasibility predicate. The primary offline benchmark uses the public NASA Solar System ISS gITF model as the station geometry source [10], with 180 area-weighted target samples and 357 viewpoint seeds used to generate SOOAs. The mesh defines target samples, visibility tests, and replay geometry; it is not a dynamics model.

The mission starts from a safe initial relative state and terminates when a coverage threshold is reached, when the planner exhausts available useful inspection actions, or when a maximum mission time is reached. For a weighted target set, the final coverage ratio is

$$C_K = \frac{\sum_{i=1}^N w_i y_{i,K}}{\sum_{i=1}^N w_i}, \quad y_{i,K} = \mathbf{1}\{\tau_{i,K} \geq \tau_{\min}\}. \quad (10)$$

The high-coverage offline benchmark uses a 70% success threshold and a 98% planning stop ratio. These two values have different roles: the former defines mission success, whereas the latter asks the planner to continue collecting coverage after success so that high-coverage behavior can be compared across baselines. The best feasible method reached 98.33% final coverage, as reported in Section VII.

IV. Problem Formulation

A. State, Visibility, and Coverage

Using the sampling convention in Section III, let the chaser state at discrete time k be

$$\mathbf{x}_k = [\mathbf{r}_k^T \quad \mathbf{v}_k^T]^T \in \mathbb{R}^6, \quad (11)$$

where $\mathbf{r}_k \in \mathbb{R}^3$ and $\mathbf{v}_k \in \mathbb{R}^3$ are the relative position and velocity in the local-vertical/local-horizontal frame of a nominal circular chief orbit. The inspected structure is represented by a finite weighted target set

$$\mathcal{T} = \{(p_i, n_i, w_i)\}_{i=1}^N, \quad (12)$$

where $p_i \in \mathbb{R}^3$ is the target point, n_i is the outward surface normal, and $w_i \geq 0$ is the inspection weight of that sample.

The camera attitude is represented by a unit boresight vector $b(q_k) \in \mathbb{S}^2$ or, equivalently, by an attitude variable q_k that determines the boresight. Illumination is not modeled as a state variable in the baseline formulation; it can be included as an additional binary feasibility predicate when mission lighting constraints are relevant.

Target i is visible at time k if range, field-of-view, incidence angle, and line-of-sight constraints all hold:

$$\rho_{\min} \leq \|p_i - \mathbf{r}_k\| \leq \rho_{\max}, \quad (13)$$

$$\arccos\left(\frac{(p_i - \mathbf{r}_k)^T b(q_k)}{\|p_i - \mathbf{r}_k\|}\right) \leq \alpha_{\max}, \quad (14)$$

$$\arccos\left(\frac{(\mathbf{r}_k - p_i)^T n_i}{\|\mathbf{r}_k - p_i\| \|n_i\|}\right) \leq \theta_{\max}, \quad (15)$$

$$\text{LOS}(\mathbf{r}_k, p_i, \mathcal{O}) = 1, \quad (16)$$

where $\alpha_{\max}$ is the half-angle field-of-view limit, $\theta_{\max}$ is the maximum incidence angle, and $\mathcal{O}$ denotes the occluding geometry. The dwell state evolves as

$$\tau_{i,k+1} = \begin{cases} \tau_{i,k} + \Delta t, & \text{if target } i \text{ is visible at } k, \\ \tau_{i,k}, & \text{otherwise.} \end{cases} \quad (17)$$

The inspection indicator is $y_{i,k} = \mathbf{1}\{\tau_{i,k} \geq \tau_{\min}\}$.

B. Dynamic Feasibility and Cost

The planner directly uses the zero-order-hold discrete HCW model in Eq. (7):

$$\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k, \quad (18)$$

where $\mathbf{u}_k \in \mathbb{R}^3$ is the commanded acceleration applied over the k th sampling interval. The chaser must satisfy the actuation and safety limits

$$\|\mathbf{u}_k\|_2 \leq u_{\max}, \quad (19)$$

$$\|\mathbf{v}_k\|_2 \leq v_{\max}, \quad (20)$$

$$d_S(\mathbf{r}_k) \geq d_{\text{safe}}, \quad (21)$$

where $d_S(\mathbf{r}_k)$ is the signed distance to the station safety envelope. A CBF-type filter can be attached later as an execution layer, but it is not part of the core mathematical problem.

The offline planner evaluates motion between SOOA terminal poses through bounded HCW rollouts. For a

transition from arc a_a to arc a_b , let π_{ab} denote the sampled state-control sequence generated over a fixed or bounded transfer duration. The terminal state is constrained to reach the seed dwell position with zero relative velocity, so each selected SOOA corresponds to a stabilized inspection action rather than a fly-through point. The transfer is feasible when every sample satisfies the input, speed, and clearance constraints and the terminal position and velocity errors remain within prescribed tolerances. We write this feasibility indicator as

$$\eta_{ab} = \mathbf{1}\{\pi_{ab} \text{ satisfies all dynamic and safety limits}\}. \quad (22)$$

The corresponding edge cost is

$$\begin{aligned} \ell_{ab} = & \lambda_v \sum_{k \in \pi_{ab}} \|\mathbf{u}_k\|_2 \Delta t + \lambda_t T_{ab} \\ & + \lambda_s \sum_{k \in \pi_{ab}} \phi(d_S(\mathbf{r}_k)) + \lambda_e e_{ab}, \end{aligned} \quad (23)$$

where T_{ab} is transfer duration, e_{ab} is terminal tracking error, including terminal position and velocity residuals, and ϕ penalizes proximity to the safety boundary. Edges with $\eta_{ab} = 0$ are either removed from the reduced finite graph or assigned a prohibitive cost in greedy baselines. The coverage mission ends when either the prescribed stop ratio is reached, the finite view budget is exhausted, or no feasible outgoing edge remains from the current terminal state.

C. Safe Observable Orbital Arc

A safe observable orbital arc is the finite-duration inspection action used by the planner. For an arc a_j over $t \in [0, T_j]$, define

$$a_j = (\mathbf{x}_j(t), \mathbf{u}_j(t), q_j(t), \Gamma_j, \Delta v_j, d_j^{\min}, \rho_j), \quad (24)$$

where $\mathbf{x}_j(t)$ is the HCW-propagated relative state, $\mathbf{u}_j(t)$ is the commanded acceleration, $q_j(t)$ is the camera attitude profile, $\Gamma_j \subseteq \mathcal{T}$ is the set of targets satisfying the range, FOV, incidence, dwell, and line-of-sight visibility predicates for the stabilized inspection action, Δv_j is the accumulated control effort, $d_j^{\min}$ is the minimum clearance, and ρ_j is the passive-safety audit margin.

The accumulated control effort and clearance are

$$\Delta v_j = \int_0^{T_j} \|\mathbf{u}_j(t)\|_2 dt, \quad (25)$$

$$d_j^{\min} = \min_{t \in [0, T_j]} d_S(\mathbf{r}_j(t)). \quad (26)$$

An arc is dynamically admissible if all sampled states satisfy the HCW propagation model, input bound, speed bound, clearance bound, and terminal error tolerance. It is observable if $\Gamma_j \neq \emptyset$ under the camera visibility model.

To audit passive safety, the uncontrolled drift from sampled states on the arc is propagated as

$$\dot{\mathbf{x}}_{\text{drift}} = A_{\text{HCW}} \mathbf{x}_{\text{drift}}. \quad (27)$$

The reported passive-safety margin is

$$\rho_j = \min_{t_k \in a_j} \min_{\tau \in [0, T_{\text{PS}}]} d_S(\mathbf{r}^{\text{drift}}(t_k + \tau)) - d_{\text{PS}}. \quad (28)$$

The arc is passively safe when $\rho_j \geq 0$. This is a planning-level audit under the sampled HCW model and the selected audit horizon; it is not a flight-certified safety guarantee.

D. Optimization Objective

The offline inspection problem is to choose a sequence of viewpoints and controls that maximizes weighted coverage while remaining dynamically feasible:

$$\begin{aligned} \min_{\mathcal{X}, \mathcal{U}, \mathcal{Q}} \quad & J_{\text{total}} \\ = & w_C(1 - C_K) + w_T K \Delta t \\ & + w_U \sum_{k=0}^{K-1} \|\mathbf{u}_k\|_2 \Delta t \\ & + w_S \sum_{k=0}^K \phi(d_S(\mathbf{r}_k)) \\ & + w_A \sum_{k=0}^K \psi(q_k) \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k, \\ & \|\mathbf{u}_k\|_2 \leq u_{\max}, \quad \|\mathbf{v}_k\|_2 \leq v_{\max}, \\ & d_S(\mathbf{r}_k) \geq d_{\text{safe}}, \\ & y_{i,k} \text{ follows the visibility and dwell model,} \\ & C_K \geq C_{\text{req}}. \end{aligned} \quad (29)$$

Here $\mathcal{X} = \{\mathbf{x}_k\}_{k=0}^K$, $\mathcal{U} = \{\mathbf{u}_k\}_{k=0}^{K-1}$, and $\mathcal{Q} = \{q_k\}_{k=0}^K$. The function $\phi(\cdot)$ penalizes proximity to the safety boundary and $\psi(q_k)$ penalizes attitude or boresight deviation if attitude is optimized explicitly. The weighted coverage ratio is

$$C_K = \frac{\sum_{i=1}^N w_i y_{i,K}}{\sum_{i=1}^N w_i}. \quad (30)$$

E. Finite Graph and Evaluation Quantities

This is a mixed-integer, nonconvex planning problem. The discrete components are target visibility, dwell satisfaction, and SOOA selection. The continuous components are HCW propagation, actuation limits, and clearance constraints. The present benchmark studies the finite approximation induced by a SOOA graph. Seed viewpoints generate terminal/dwell camera poses, but the accepted graph elements are finite-duration HCW-feasible inspection arcs.

Let $\mathcal{A} = \{a_j\}_{j=1}^M$ be the accepted SOOA library and let $\Gamma(a_j) \subseteq \mathcal{T}$ be the target subset visible from arc a_j . For a maximum of L selected arcs, the finite planning problem is to choose an ordered sequence

$$\sigma = (a_{\sigma_1}, \dots, a_{\sigma_L}) \quad (31)$$

that satisfies feasible-transfer constraints and covers enough target weight. One representative objective is

$$J_{\text{graph}}(\sigma) = \sum_{\ell=1}^L \ell(a_{\sigma_{\ell-1}}, a_{\sigma_{\ell}}) + \lambda_v \sum_{\ell=1}^L \Delta v(a_{\sigma_{\ell}}) + \lambda_L L + \lambda_C \sum_{i=1}^N w_i(1 - c_i). \quad (32)$$

where a_{σ_0} denotes the initial state action and

$$c_i = \mathbf{1} \left[i \in \bigcup_{\ell=1}^L \Gamma(a_{\sigma_{\ell}}) \right]. \quad (33)$$

The coverage constraint is

$$\sum_{i=1}^N w_i c_i \geq C_{\text{req}} \sum_{i=1}^N w_i. \quad (34)$$

This finite graph is the object on which the exact optimality statement in Section E is made. The implementation evaluated in Section VII uses a scalable deterministic safe-arc coverage solver on the same graph.

For reporting, the primary quantities are cumulative control effort, minimum clearance, peak requested input, mission time, and selected-SOOA count:

$$\Delta v_{\text{cum}} = \sum_k \|\mathbf{u}_k\|_2 \Delta t, \quad (35)$$

$$d_{\min} = \min_k d_S(\mathbf{r}_k), \quad (36)$$

$$u_{\max, \text{req}} = \max_k \|\mathbf{u}_k\|_{\text{req}, 2}. \quad (37)$$

These quantities are used later to compare planners on the same mission geometry and relative dynamics model.

V. Offline Planner

A. Decomposition

The offline solver operates on the finite graph defined in Section IV. It decomposes inspection into target sampling, viewpoint-seed construction, SOOA generation, visibility-matrix construction, HCW feasibility auditing, finite-graph optimization, trajectory rollout, and camera-attitude export. The implementation has two modes. The certified mode solves a reduced SOOA graph exactly by dynamic programming over selected-arc masks and the terminal arc. The scalable mode uses deterministic safe-arc coverage selection followed by CW-cost tour ordering for the full ISS-mesh benchmark. Both modes use the same camera visibility matrix and the same HCW transfer feasibility audit, so the certificate and the scalable results refer to the same discretized mission model.

B. Viewpoint Seeds and SOOA Library

For each surface target, viewpoint seeds are generated on offset shells and accepted only when they satisfy basic range, field-of-view, incidence-angle, line-of-sight, and clearance constraints. Each seed then induces a transfer-to-dwell SOOA whose terminal camera pose is used for

visibility in the current implementation. Each SOOA a_j stores a safety margin, transfer metrics, passive-drift audit fields, and a precomputed set of visible targets. The resulting binary visibility relation can be written as

$$A_{ij} = 1 \iff \text{target } i \text{ is visible from SOOA } a_j. \quad (38)$$

In the high-coverage offline benchmark, 357 candidates are generated for 180 mesh target samples.

C. SOOA Selection

The scalable solver selects SOOAs on the accepted arc graph and then orders the selected sequence using CW transfer costs. At each selection step, the planner balances new weighted coverage against the HCW transfer cost from the current predicted state. For arc a_j , let ΔC_j be the newly gained weighted coverage and let $J(x, a_j)$ be the rollout cost defined by the edge model in Section IV. The selection stage uses the score

$$s_j = \frac{\Delta C_j}{1 + J(x, a_j)} + w_m \min\{\hat{d}_{\min}(x, a_j), d_{\max}\}, \quad (39)$$

then updates the predicted state with the CW rollout to the selected SOOA terminal state. The resulting ordered sequence is then propagated forward so that later choices depend on the orbital state reached by earlier ones. The planner records the selected SOOA identifier, seed-view identifier, new target count, area gain, cumulative coverage, score, transfer Δv , dynamic transfer cost, minimum clearance, transfer tracking error, and feasibility flag for each selected inspection action. The planning loop terminates when the cumulative coverage reaches the prescribed stop ratio, when the view budget is exhausted, or when no feasible outgoing edge remains from the current state.

D. CW Transfer Generation

After an arc is selected, the planner computes a finite-horizon HCW terminal transfer to the selected dwell pose. For the fixed integration grid, the linear HCW map is used to solve the minimum-energy sequence of piecewise constant accelerations whose terminal position equals the SOOA dwell position and whose terminal relative velocity is zero. The transfer is then sampled forward, and the edge is marked feasible only when terminal position error, terminal speed, peak input, speed, and clearance satisfy the specified bounds. This makes the selected SOOAs stabilized inspection actions rather than fly-through camera markers. Because each segment is generated independently on the discrete HCW grid, the reported trajectory is physically valid but piecewise smooth rather than a single globally optimized spline. The plotted line is a stitched sequence of rest-to-rest transfers, so corners can appear at viewpoint transitions even though each segment satisfies the HCW dynamics model.

The exported trajectory includes position, velocity, acceleration command, camera boresight, yaw, pitch, and

Algorithm 1 SOOA-based orbital coverage planner

Require: Structure geometry $\mathcal{S}$, initial state x_0 , camera model, HCW model, safety limits, C_{req} , C_{stop}
Ensure: Planned trajectory $\mathcal{P}$, attitude stream, SOOA sequence, and inspection metrics

- 1: Sample area-weighted surface targets $\mathcal{T} = \{(p_i, n_i, A_i)\}_{i=1}^N$
- 2: Generate viewpoint and attitude seeds around the structure
- 3: For each seed, generate a finite-duration terminal/dwell candidate arc under the HCW model
- 4: Reject arcs violating terminal error, input, speed, clearance, or passive-audit constraints
- 5: Evaluate range, FOV, incidence, dwell, and LOS visibility for accepted arcs
- 6: Construct the SOOA library $\mathcal{A}$ with visibility sets $\Gamma(a_j)$, costs Δv_j , clearances $d_j^{\min}$, and margins ρ_j
- 7: Initialize covered set $\mathcal{C} \leftarrow \emptyset$, selected sequence $\sigma \leftarrow \emptyset$, and predicted state $x \leftarrow x_0$
- 8: **while** $C(\mathcal{C}) < C_{\text{stop}}$ and the arc budget is not exhausted **do**
- 9: Compute new weighted coverage gain ΔC_j for feasible outgoing SOOAs
- 10: Score each arc using coverage gain, transfer cost, control effort, and safety margin
- 11: Select $a^* = \arg \max_j s_j$
- 12: Append a^* to σ and update the predicted terminal state
- 13: Update covered set $\mathcal{C}$ using $\Gamma(a^*)$
- 14: **end while**
- 15: Roll out and audit the selected SOOA sequence under HCW dynamics
- 16: Export trajectory, control, attitude, FOV, coverage, safety, and planner logs
- 17: **return** $\mathcal{P}$, attitude stream, SOOA sequence, and metrics

quaternion samples, so the planned path specifies both where the chaser flies and how the inspection camera is pointed. Edges that violate feasibility checks are penalized or rejected depending on the baseline method. Thus, the same visibility gain can be accepted or rejected depending on whether the terminal HCW transfer remains inside the control and clearance envelope. The high-coverage benchmark evaluates the scalable graph solver rather than claiming a global mixed-integer optimum for the continuous mission.

E. Certified Finite-Graph Solver

Let $\mathcal{A} = \{a_j\}_{j=1}^M$ denote the finite SOOA set. Each arc a_j observes a subset of targets $\Gamma(a_j) \subseteq \mathcal{T}$, and each directed transition (a_a, a_b) has a precomputed HCW cost $\ell_{ab} \geq 0$ and feasibility flag $\eta_{ab} \in \{0, 1\}$. The discretized mission is therefore a finite optimization problem on $\mathcal{G} = (\mathcal{A}, \mathcal{E})$, where $\mathcal{E}$ contains only feasible transfers.

Algorithm 2 Certified reduced-SOOA-graph dynamic program

Require: SOOA graph $\mathcal{G}$, visibility sets $\Gamma(a_j)$, edge costs ℓ_{jk} , feasibility flags η_{jk} , C_{req} , L_{max}
Ensure: Minimum-cost feasible sequence on the reduced graph, or no certificate

- 1: Precompute target masks for all SOOAs
- 2: Initialize $D(\emptyset, s) \leftarrow 0$ at the initial spacecraft state s
- 3: **for** $\ell = 0, \dots, L_{\text{max}} - 1$ **do**
- 4: **for all** reachable states (B, j) with $|B| = \ell$ **do**
- 5: **for all** $k \notin B$ with $\eta_{jk} = 1$ and new visible targets **do**
- 6: $B^+ \leftarrow B \cup \{k\}$
- 7: $D(B^+, k) \leftarrow \min\{D(B^+, k), D(B, j) + \ell_{jk}\}$
- 8: Store predecessor if the value improves
- 9: **end for**
- 10: **end for**
- 11: **end for**
- 12: Select the least-cost state whose weighted coverage exceeds C_{req}
- 13: Reconstruct the predecessor chain and report certificate status

Let $\Sigma_{L_{\text{max}}}$ be the set of all ordered candidate sequences of length at most L_{max} that start from the initial node and use only feasible edges. For $\sigma = (a_{\sigma_1}, \dots, a_{\sigma_L})$, define

$$c_i(\sigma) = \mathbf{1}\{i \in \cup_{\ell=1}^L \Gamma(a_{\sigma_\ell})\}, \quad (40)$$

$$J_{\text{disc}}(\sigma) = \sum_{\ell=1}^L \ell_{\sigma_{\ell-1}, \sigma_\ell}. \quad (41)$$

The exact finite inspection problem is

$$\begin{aligned} \sigma^* &= \arg \min_{\sigma \in \Sigma_{L_{\text{max}}}} J_{\text{disc}}(\sigma) \\ \text{s.t.} \quad &\sum_{i=1}^N w_i c_i(\sigma) \geq C_{\text{req}} \sum_{i=1}^N w_i. \end{aligned} \quad (42)$$

This sequence problem can be encoded as a mixed-integer program with stage-selection and edge-consistency variables. In the implementation, the certified mode solves it directly by dynamic programming. A state is (B, j) , where $B \subseteq \mathcal{A}$ is the selected-arc set and j is the last selected arc. The value $D(B, j)$ stores the minimum HCW dynamic cost of any feasible sequence that visits exactly B and terminates at j . The recursion is

$$D(B \cup \{k\}, k) = \min_{j \in B, \eta_{jk}=1} [D(B, j) + \ell_{jk}], \quad (43)$$

with initialization from the initial spacecraft state. The solver returns the lowest-cost state whose accumulated weighted coverage satisfies the required ratio before the view budget is exceeded. If all reachable states within the budget are expanded before the time and expansion limits are reached, the returned sequence is certified optimal on the reduced graph.

THEOREM 1 (Global optimality on an exhausted reduced SOOA graph) *For a fixed finite SOOA set $\mathcal{A}$, fixed visibility sets $\Gamma(a_j)$, fixed edge costs ℓ_{ab} , fixed feasibility indicators η_{ab} , required weighted coverage C_{req} , and finite horizon L_{max} , the dynamic-programming solver described above returns a minimum cost feasible inspection sequence for the discretized mission graph whenever it terminates after expanding all reachable states within the horizon.*

Sketch:

The sequence set $\Sigma_{L_{\text{max}}}$ is finite because both the candidate set and maximum sequence length are finite. Removing dynamically infeasible edges only reduces this set. The objective assigns a finite scalar value to every feasible sequence, and the coverage constraint selects a finite subset of those sequences. If that subset is nonempty, a minimum is attained. The dynamic program stores the minimum cost for every reachable pair (B, j) by Bellman's principle of optimality: any optimal sequence ending at j through selected set B must contain an optimal predecessor sequence ending at some $a \in B \setminus \{j\}$. Exhausting all reachable states of cardinality $\leq L_{\text{max}}$ therefore evaluates every feasible ordered sequence up to duplicate prefixes represented by the same state. Taking the least-cost state that satisfies the weighted coverage constraint gives the global optimum on the discretized graph. ■

The theorem is deliberately discrete. It does not claim global optimality for the original continuous viewpoint space, nor does it claim that the scalable solver used in the full ISS benchmark is globally optimal. Rather, it identifies the finite problem for which a reduced-graph oracle is meaningful. In the reported oracle smoke case, the exact solver uses eight SOOA nodes and 13 visible target IDs, expands 39 feasible states, selects two arcs, reaches the required 25% coverage, and returns certificate status "optimal". This small case verifies the finite objective and implementation; it is not a mission-level coverage result.

F. Planner Outputs

The planner outputs a timestamped reference trajectory

$$\mathcal{P} = \{(t_k, r_k^{\text{ref}}, v_k^{\text{ref}}, u_k^{\text{ref}})\}_{k=0}^K. \quad (44)$$

For the reported proposed tour, $\mathcal{P}$ contains 945 trajectory samples over 1890 s with a 90 s transfer duration and 2 s integration step. The corresponding attitude stream also contains 945 samples. The trajectory and attitude stream are subsequently available to the ROS stack for replay, logging, RViz inspection, and Gazebo visualization. In the paper, these outputs are algorithmic outputs; ROS and Gazebo form the verification environment.

VI. Execution and Experimental Protocol

The experiments are organized around the offline planner. The ROS 2 and Gazebo layers are used after planning to replay the saved trajectory, instantiate the camera geometry, and check integration consistency. They are not used as the source of orbital dynamics or as an additional optimizer.

A. Benchmark Definition

The experiments separate certification from scalability. The certification case uses a reduced ISS-mesh instance with 24 sampled surface targets, 16 generated viewpoint seeds, and an exact solver limit of eight reduced SOOA nodes. The exact dynamic program minimizes HCW dynamic cost subject to feasible transitions, a four-arc budget, and a 25% coverage requirement. This case is small enough to exhaust the reachable finite graph and therefore tests the reduced-graph oracle implemented in Algorithm 2.

The primary full-scale quantitative benchmark uses the open NASA ISS glTF mesh geometry [10], CW/HCW dynamics, 180 area-weighted mesh targets, 357 viewpoint seeds, a 70% coverage success threshold, and a 98% planning stop ratio. This benchmark uses the scalable SOOA/CW tour solver rather than the exact dynamic program. It compares the proposed tour with coverage-greedy, safe coverage-greedy, nearest-viewpoint, fuel-focused, CW-NBV, and random-safe baselines. All methods use the same SOOA seed set, visibility model, transfer duration, input limit, keep-out distance, and termination logic. A run terminates when the stop ratio is reached, when the SOOA budget is exhausted, or when no feasible outgoing transfer remains. Older Monte Carlo outputs are retained only as software regression checks; the manuscript's quantitative claims are based on the certification case, the deterministic high-coverage mesh comparison, and the validation matrix below.

To test whether the base-case conclusion is an artifact of one discretization, we also run the sensitivity and ablation matrix in Table II. The matrix is generated by the same experiment runner and writes the same CSV/JSON schema as the base benchmark: per-method coverage, cumulative Δv , peak requested input, clipped-step ratio, minimum clearance, selected-SOOA count, mission duration, and planning time.

B. Online Replay Interface

The offline planner exports

$$\mathcal{P} = \{(t_k, r_k^{\text{ref}}, v_k^{\text{ref}}, u_k^{\text{ref}}, q_k^{\text{ref}})\}_{k=0}^K, \quad (45)$$

where q_k^{ref} defines the camera attitude. Online execution tracks this sequence using the current state estimate $\hat{x}_k$ and a nominal law

$$u_k^{\text{nom}} = u_k^{\text{ref}} + K_x(\hat{x}_k - x_k^{\text{ref}}), \quad (46)$$

TABLE II: Validation matrix for sensitivity and ablation studies.

Study	Sweep
Target density	$N = 180, 500$ comparison; $N = 1000$ proposed-only stress case
Candidate density	candidate stride $\in \{1, 2, 3\}$
Initial condition	three safe initial relative states
Safety and duration	keep-out distance $\in \{1, 2, 4\}$ m; segment duration $\in \{70, 90, 120\}$ s
Ablation	remove transfer cost, clearance filtering, input checking, or coverage weighting

with K_x chosen for the HCW linearization. A runtime safety layer can pass the nominal command through the quadratic-program CBF filter

$$\begin{aligned}
 u_k^{\text{exec}} = \arg \min_{u \in \mathbb{R}^3} & \quad \|u - u_k^{\text{nom}}\|_2^2 \\
 \text{s.t.} & \quad h(x_k) \geq 0, \\
 & \quad \dot{h}(x_k, u) + \alpha h(x_k) \geq 0, \\
 & \quad \|u\|_2 \leq u_{\max},
 \end{aligned} \tag{47}$$

where $h(x_k) = d_S(r_k) - d_{\text{safe}}$ and $\alpha > 0$ is a barrier gain. A new offline plan is requested if the tracking error exceeds a tolerance, the barrier filter remains active beyond a prescribed duration, the executed coverage falls behind its predicted envelope, or the state estimator reports loss of consistency. This layer defines the execution interface; the planner itself remains offline.

C. ROS/Gazebo Verification

The ROS execution run records odometry, nominal control, safe control, coverage map, mission events, safety status, planner events, camera attitude, and field-of-view topics. The replay node reconstructs the camera frame from the saved attitude stream, publishes the boresight marker, and generates the camera frustum from the same field-of-view parameters used during planning. This verifies that the visual model used in the offline visibility predicate can be instantiated consistently at runtime.

Gazebo Harmonic is used to visualize the station and chaser after the ROS state and camera streams have been reconstructed. The validation artifacts include preview frames and videos from chaser-follow and whole-structure views with a camera-frustum overlay. Gazebo does not determine the spacecraft state in these experiments. The chaser pose is driven from the ROS state stream, so the dynamics node remains the source of truth.

VII. Results

A. Offline Planning Performance

Table III summarizes the proposed SOOA tour in the high-coverage mesh experiment. The planner reached 98.33% coverage over 180 surface targets, exceeding the 70% success threshold and passing the 98% planning

TABLE III: Best feasible offline planning metrics from the high-coverage mesh experiment.

Metric	Value
Method	Proposed SOOA tour
Dynamics model	CW/HCW
Geometry model	NASA ISS gITF mesh
Target count	180
Viewpoint seeds	357
Selected SOOAs	21
Trajectory samples	945
Mission duration	1890.0 s
Final coverage ratio	0.9833
Inspectable coverage ratio	0.9833
Coverage threshold	0.7000
Coverage success	Yes
Total planned Δv	21.5921 m/s
Δv per coverage	21.9581 m/s
Maximum speed	0.8898 m/s
Minimum clearance	9.9107 m
Passive audit horizon	300.0 s
Minimum passive audit margin	1.5091 m
Passive audit passed	Yes
RMS transfer tracking error	19.9695 m
Peak requested input	0.0388 m/s ²
Clipped step ratio	0.0000
Planning time	5.6254 s
Feasible trajectory	Yes

TABLE IV: Reduced-SOOA-graph oracle check. The exact dynamic program minimizes HCW dynamic cost subject to feasible transitions, a four-arc budget, and a 25% coverage requirement.

Method	Coverage	Dynamic cost
Exact DP oracle	0.2500	26.7682
Proposed scalable	0.2917	27.0924
Safe coverage greedy	0.2917	30.3664

stop target. Coverage is reported together with cumulative Δv , minimum clearance, peak requested input, and clipping ratio because these quantities determine whether an inspection sequence is dynamically plausible. These results establish the scalable full-graph comparison only; the sensitivity and ablation studies in Table II test which parts of this ranking persist under denser target sampling and perturbed mission conditions. The selected SOOA sequence also passes the reported passive-drift audit with a minimum margin of 1.509 m over a 300 s uncontrolled HCW drift horizon.

Table IV reports a separate reduced-graph oracle check. The exact dynamic program used eight SOOA nodes from a 16-seed mesh instance, covered 13 visible target IDs, expanded 39 feasible graph states, and returned certificate status “optimal”. The certified sequence selected two arcs, reached the required 25% raw coverage, and remained within the input and clearance limits. This is intentionally not a headline coverage result; it verifies that the finite SOOA objective and implementation can be exhausted on a small graph.

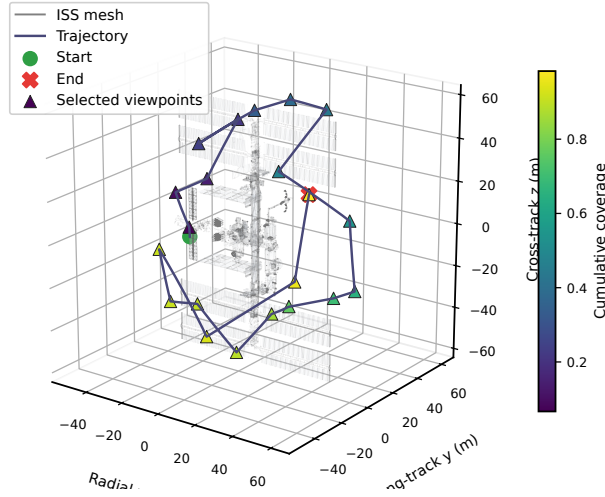


Fig. 1: Proposed SOOA tour overlaid on the open NASA ISS glTF mesh model. The selected SOOA terminal poses are colored by cumulative coverage, and the trajectory is shown as the planned chaser path in LVLH coordinates.

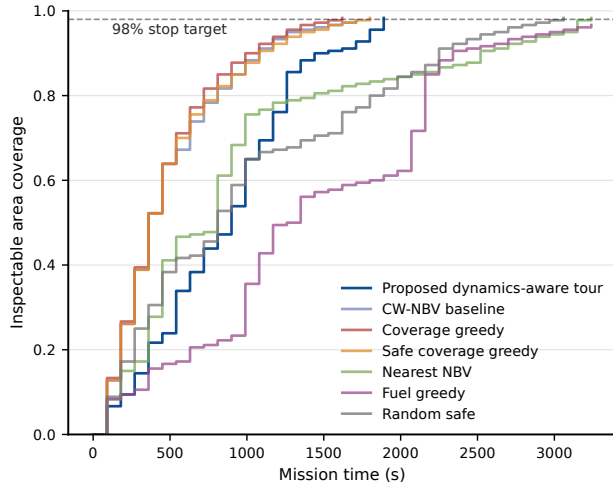


Fig. 2: Coverage evolution for the compared offline methods in the high-coverage mesh experiment.

Figure 1 overlays the planned trajectory on the same open NASA ISS glTF mesh model used for target sampling and replay visualization [10]. The figure is generated from the benchmark data, so the displayed path and selected SOOA terminal poses share the same LVLH coordinate frame as the mesh. The path is a stitched sequence of discrete rest-to-rest HCW transfers, so it is dynamically valid but not drawn as a single global spline.

Figure 2 reports the corresponding coverage evolution for the deterministic mesh comparison. All curves are generated from the archived high-coverage experiment and use the same candidate set, visibility model, transfer constraints, and termination logic.

Figure 3 compares the baseline trajectories as method-wise small multiples. The proposed SOOA tour is shown separately in Fig. 1, where the selected SOOA terminal poses and full mesh context are easier to inspect. The

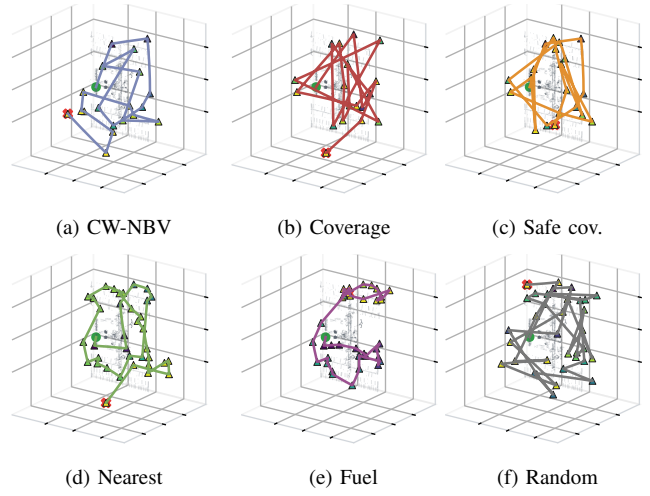


Fig. 3: Baseline trajectory comparison over the same ISS mesh silhouette. The proposed SOOA tour is shown separately in Fig. 1.

shared mesh silhouette, viewpoint, and axis limits make baseline path geometry comparable without repeating the proposed trajectory. Figure 4 reports the full comparison in terms of final coverage and cumulative Δv , and it marks dynamically infeasible runs. In this representation, the proposed tour is the lowest-energy feasible method that reaches the high-coverage stop target in the reported base case. The coverage-greedy and nearest-viewpoint runs can reach similar coverage only by violating input or clearance constraints, whereas the safe-coverage and random-safe baselines remain feasible but require substantially more cumulative Δv . The fuel-focused baseline uses low input effort but stops below the high-coverage target. The baseline trajectory panels are intentionally unsmoothed between selected SOOAs. The visible corners therefore reflect terminal HCW transfer segments, not a rendering artifact.

Table V reports execution and solver quantities that are not shown in Fig. 4. Rest-to-rest terminal transfers remove the earlier fly-through artifact in which a myopic safe baseline could terminate after one view. Safe-coverage-greedy now reaches the same final coverage as the proposed tour, but it requires more than twice the cumulative Δv . Coverage-greedy reaches the target coverage with fewer views, but it violates input and clearance checks. Nearest-viewpoint selection also reaches high coverage, but its clearance violation makes the resulting tour dynamically infeasible.

Figure 5 summarizes the sensitivity and ablation matrix. Increasing the target count from 180 to 500 preserves the main ranking: the proposed tour remains feasible and reaches the 98% stop target with 21.38 m/s cumulative Δv , whereas safe-coverage-greedy requires 47.08 m/s and coverage-greedy remains infeasible. The coarser $N = 1000$ proposed-only stress case remains feasible but reaches 76.5% coverage because the candidate

TABLE V: Complementary offline comparison quantities not encoded by the coverage-effort plot.

Method	SOOAs	Duration (s)	Peak input (m/s^2)	Clipped steps (%)	Planning time (s)	Feasible
Proposed	21	1890	0.0388	0.0	5.63	Yes
CW-NBV baseline	19	1710	0.0581	0.0	24.40	Yes
Coverage greedy	18	1620	0.0849	10.5	1.21	No
Safe coverage greedy	20	1800	0.0581	0.0	0.97	Yes
Nearest NBV	36	3240	0.0528	0.0	4.82	No
Fuel greedy	36	3240	0.0368	0.0	4.68	Yes
Random safe	34	3060	0.0588	0.0	6.50	Yes

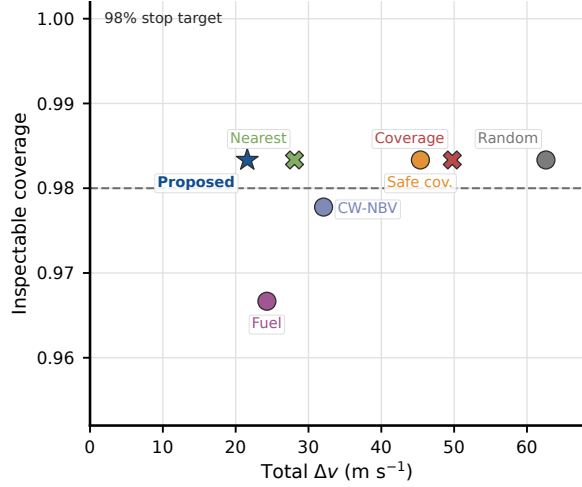


Fig. 4: Coverage-effort-feasibility summary for the full offline comparison. The dashed line in the summary panel marks the 98% stop target. A star denotes the proposed method shown in Fig. 1; circles denote feasible baselines, and crosses denote methods that fail the input, speed, or clearance checks.

stride is increased to keep mesh visibility computation tractable; this identifies candidate-density coupling as a remaining scalability issue rather than hiding it.

The initial-condition sweep over three safe starting states keeps the proposed tour feasible with final coverage 98.33%, cumulative Δv between 20.19 and 23.09 m/s , and positive clearance. Candidate-density sweeps show that reducing candidate count from 357 to 120 still reaches at least 97.2% coverage but reduces clearance to 0.73 m in the proposed tour. Transfer duration is the clearest feasibility boundary: a 70 s segment requests 0.061 m/s^2 , slightly above the 0.060 m/s^2 limit, and is therefore audited as infeasible; 90 s and 120 s segments are feasible.

The ablation study provides direct evidence for the algorithmic components. Removing transfer cost increases cumulative Δv and peak input while retaining coverage. Removing the input-feasibility check produces high coverage but violates the input limit. Removing clearance filtering changes the selected set and reduces final coverage to 90.0% in this benchmark. Unweighted coverage is nearly identical here because the sampled ISS targets have similar effective weights, but it remains an explicit

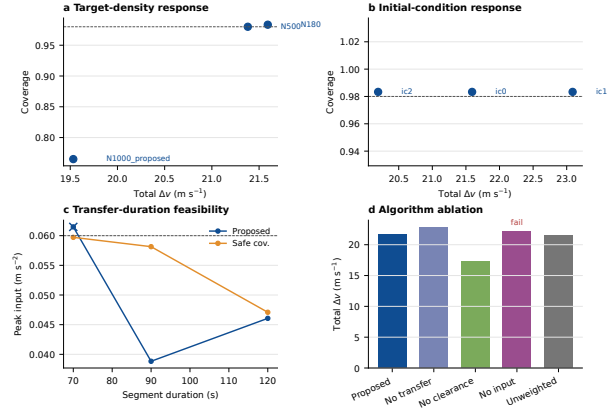


Fig. 5: Sensitivity and ablation results generated by the offline validation matrix. Circles denote feasible runs and crosses denote runs that fail the full feasibility audit. The dashed lines mark the 98% stop target or the 0.060 m/s^2 input limit.

ablation because area weighting becomes important for nonuniform meshes.

B. Camera Geometry and Replay Verification

The high-coverage experiment exports trajectory and camera-attitude streams for every compared method. For the proposed tour, each stream contains 945 samples. The camera orientation data replay the planned path with the same boresight alignment used in the visibility predicate. The trajectory replay node uses these data to publish odometry, path markers, field-of-view markers, and station mesh markers for RViz and Gazebo validation.

Figures 6 and 7 make the camera model visible in two ways. The first reconstructs representative field-of-view frusta from selected SOOA terminal poses and boresight vectors. The second reports the incremental surface coverage contributed by the selected SOOA sequence. Together, they show that saved attitudes instantiate the same FOV geometry used by the visibility predicate and that the selected inspection actions contribute to the final coverage.

The ROS execution run reconstructs the planned state, attitude, and camera frame from the saved logs, broadcasts the same FOV geometry used by the planner, and records

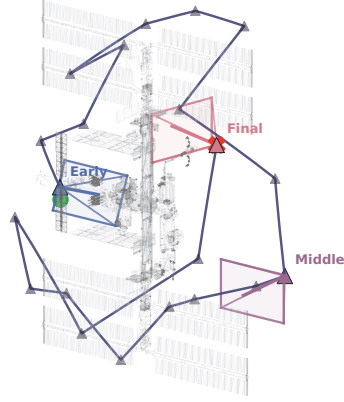


Fig. 6: Camera-model verification through reconstructed field-of-view geometry overlaid on the proposed SOOA tour. The frusta and boresight rays are reconstructed from saved camera poses for representative early, middle, and final selected SOOAs, verifying that the logged attitude stream instantiates the visibility model used by the planner and by ROS replay in the same LVLH frame as the ISS mesh.

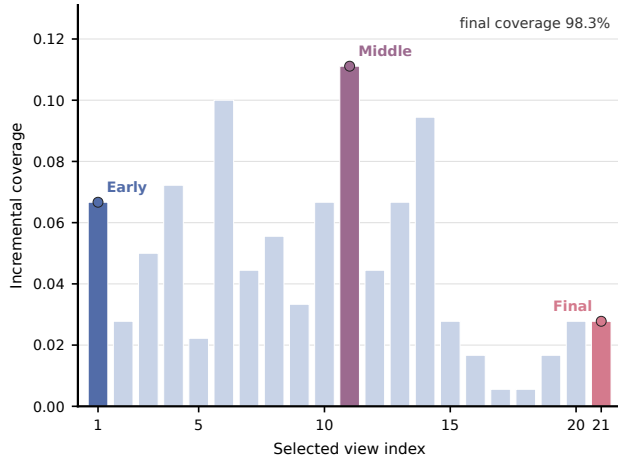


Fig. 7: Incremental surface coverage contributed by the selected SOOAs. Bars show per-arc coverage gain in the selected SOOA sequence, with the same representative early, middle, and final arcs highlighted as in Fig. 6.

trajectory, control, coverage, mission-event, safety, and planner topics. This verifies that the visual model, attitude stream, and geometric visibility logic remain consistent after integration.

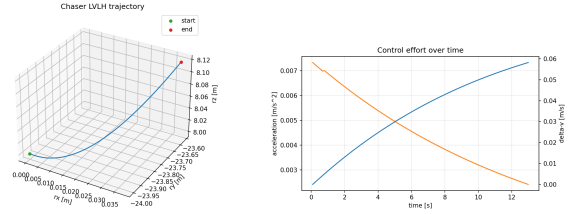


Fig. 8: ROS replay log verification for the planned SOOA execution. Left: replayed chaser trajectory in the same LVLH frame used by the offline planner. Right: logged control effort over replay time. These panels verify that the saved trajectory and command streams are available to the ROS logging and visualization stack; orbital dynamics claims remain based on the HCW planner logs.



Fig. 9: Gazebo Harmonic validation frames. Left: chaser-follow camera view. Right: full ISS visual context with chaser trajectory replay. Quantitative coverage, safety, and control claims come from recorded planner logs, not Gazebo physics.

VIII. Discussion and Limitations

The main result is that coverage and orbital feasibility cannot be separated without changing the mission outcome. In the reported benchmark, the proposed SOOA tour reaches 98.33% coverage with positive clearance, the lowest Δv among feasible methods that reach the high-coverage stop target, and a comparatively low peak requested input. The coverage-greedy method reaches the same coverage but requires acceleration beyond the admissible input limit, while nearest and random-safe selection also become infeasible because of clearance or input violations. Coverage alone is therefore not an adequate objective for orbital inspection.

The sensitivity matrix strengthens this conclusion but also shows its boundary. The proposed planner remains feasible under the $N = 500$ target density, three tested initial states, candidate strides from 1 to 3, and keep-out margins from 1 to 4 m. However, the coarser $N = 1000$ stress case falls below the 98% stop target because candidate density was reduced for runtime, and the 70 s transfer-duration case violates the input limit. Thus, the method is not simply “robust” in an unqualified sense. It is robust over the completed moderate sweeps, but high-density mesh coverage requires better candidate generation, visibility acceleration, or a larger view budget.

The benchmark still uses a downsampled line-of-sight representation for occlusion checking and a CW/HCW translational model. These choices are appropriate for a reproducible algorithmic baseline, but the validation

matrix shows that dense target sets quickly make line-of-sight and transfer evaluation expensive. A flight-oriented planner would require accelerated mesh queries, uncertainty-inflated keep-out volumes, camera slew limits, illumination constraints, state-estimation error, plume constraints, and actuator saturation studies. The reported RMS transfer tracking error is also a planner-side rollout metric rather than a closed-loop flight validation error.

The optimality statement is similarly bounded. The implemented dynamic program certifies the minimum-cost sequence only on the fixed reduced candidate graph whose reachable states are fully expanded. The reported high-coverage ISS experiment uses a scalable deterministic solver on the same graph model, so its evidence is empirical rather than a global certificate for the full graph or for the original continuous viewpoint space. This split is intentional: the reduced case verifies the optimization statement, whereas the full benchmark tests whether the same dynamics-aware modeling choices remain useful at ISS mesh scale.

Gazebo validation should be interpreted conservatively. The videos and frames verify that the visual model, chaser model, camera view, and pose-following interfaces work in the integrated stack. They do not validate orbital mechanics, prove coverage, or certify collision avoidance. Those quantitative claims come from the HCW dynamics, coverage, trajectory, attitude, safety, and planner logs.

IX. Conclusion

This paper introduced safe observable orbital arc planning for autonomous inspection of orbital structures. The planner constructs finite-duration inspection actions that are simultaneously HCW-feasible, FOV-valid, coverage-informative, and safety-audited instead of checking dynamics after geometric coverage has been chosen. The reduced-SOOA certification case showed that the implemented dynamic program can return a true finite-graph optimum: with eight candidate arcs and 13 visible target IDs, it expanded 39 feasible states and selected a minimum-cost two-arc tour satisfying the 25% coverage requirement. On the full ISS-mesh benchmark, the scalable solver generated a feasible 1890 s trajectory with 21 selected inspection actions, 21.5921 m/s cumulative Δv , 9.9107 m minimum clearance, and 98.33% final coverage over 180 surface targets. Compared with coverage-greedy, safe-coverage, nearest-viewpoint, fuel-focused, CW-NBV, and random-safe baselines, the proposed solver reached the target coverage with the lowest cumulative Δv among feasible high-coverage methods and avoided the input and clearance violations observed in simpler high-coverage selection rules. ROS 2 and Gazebo provided replay and camera-geometry verification, while the planner and logged HCW trajectory remained the basis of the quantitative claims.

Additional sensitivity and ablation runs show that the proposed planner remains feasible for the tested $N = 500$ target-density case, three initial states, three candidate

densities, and keep-out margins from 1 to 4 m. They also expose real limits: a 70 s transfer segment violates the input bound, and a coarse $N = 1000$ stress case falls below the high-coverage stop target. Future work should therefore focus on accelerated mesh visibility, denser candidate generation, broader Monte Carlo cases, larger certified graph instances, and uncertainty-aware keep-out, camera slew, illumination, and actuator constraints. Higher-fidelity validation, including an optional Basilisk backend, should be added only after the HCW baseline remains reproducible.

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